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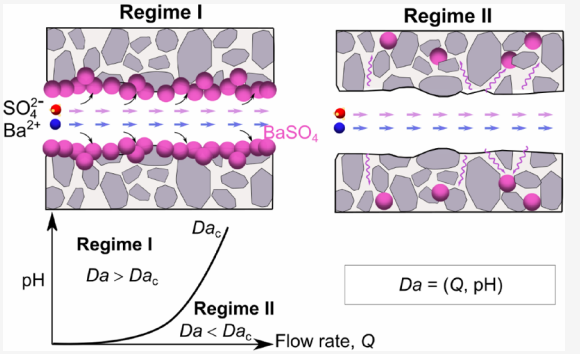
Controls of the Nucleation Rate and Advection Rate on Barite Precipitation in Fractured Porous Media

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ABSTRACT: Mineral precipitation is ubiquitous in natural and engineered environments, such as carbon mineralization, contaminant remediation, and oil recovery in unconventional reservoirs. The precipitation process continuously alters the medium permeability, thereby influencing fluid transport and subsequent reaction kinetics. The diversity of preferential precipitation zones controls flow and transport efficiency as well as the capacity of mineral sequestration and immobilization. Taking barite precipitation as an example, previous studies have examined this process in porous and/or fractured media, but pore-scale mechanisms under varying flowing and geochemical conditions remain unexplored. In this study, we conducted real-rock microfluidic experiments to investigate the precipitation dynamics within a fractured porous system. Direct observations of the evolution of the porous structure and flow channel and quantifications of barite precipitation dynamics using X-ray diffraction (XRD) and scanning electron microscopy with energy-dispersive X-ray spectroscopy (SEM-EDS), revealed two distinct precipitation regimes: precipitation on the fracture surface (regime I) and precipitation in the alteration zone (regime II). Through theoretical analysis of the rate of advection and nucleation, we defined a dimensionless number *Da* above which regime I occurs and regime II prevails otherwise. At the large *Da* number, when the precipitation rate is large compared with the flow rate, precipitation on the fracture surface is favored. As the precipitation regimes are expected to impact differently the permeability of the fractured porous media, the mass transfer across matrix and fractures, and the spatial distributions of coprecipitated contaminants, our work sheds light on accurately modeling reactive transport in fractured porous media across diverse applications.



INTRODUCTION

Mineral precipitation in permeable media is a widely observed phenomenon with important implications for various environmental and industrial processes, such as carbon mineralization,[1−3] contaminant remediation,[4−7] and energy harvesting.[8−10] The precipitation processes continuously alter the permeability of the media, thereby influencing fluid transport and subsequent reactions.[11−13] The spatial patterns of precipitation controls the mineral sequestration capacity and flow/transport efficiency of the media.[14−17] Understanding the mineral precipitation process is essential for predicting the long-term behavior of contaminants in subsurface environments and designing effective remediation strategies. For instance, previous experimental studies have shown that barite ($BaSO_4$), a mineral of interest in many energy and environmental systems, can precipitate on the surface of shale fracture and/or within the shale matrix, depending on the geochemical and flowing conditions.[18−21] This study chooses the barite-shale model system to investigate the formation of precipitation zones under various conditions to provide insights crucial for developing effective strategies for subsurface resource management and environmental protection.

Previous research has focused on mineral precipitation in solution systems[22−25] and on rock or fracture surfaces[26,27] and improved our understanding of the formation of preferential precipitation zones influenced mainly by pH and flow rate. It is widely recognized that under nonflowing conditions, precipitation is promoted by higher pH (pH > 10) and suppressed at pH < 3.[28] Li et al. reported distinct precipitation zones under various pH conditions in a barite-shale system. In shale systems with higher carbonate contents, the increase in pH due to carbonate dissolution facilitates mineral precipitation within microfractures and pore spaces. For shales with lower carbonate contents, little precipitation occurs at the shale−fluid interface.[18,29] Core-scale experiments under reservoir conditions by Vankeuren et al. also highlighted a similar dependence of mineral precipitation on pH.[27]

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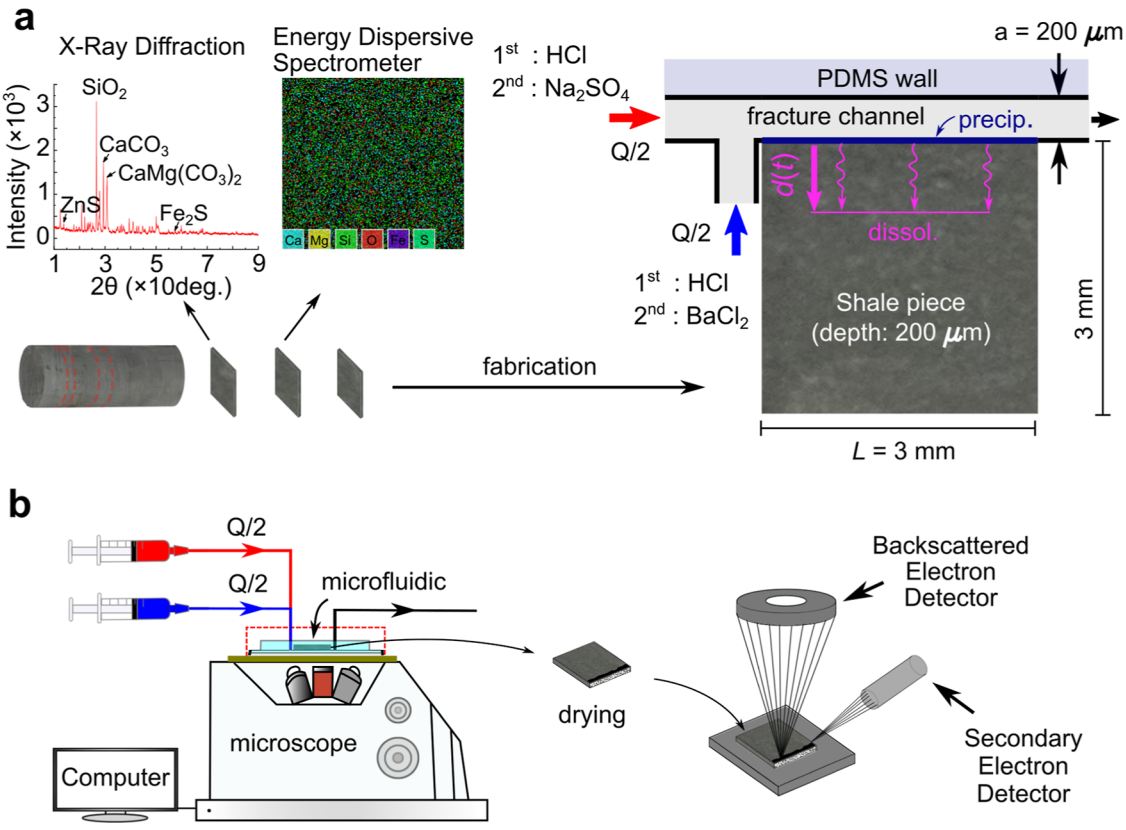


Figure 1. Shale microfluidic flow cell and experimental setup. (a) Mineral composition and sample preparation. The shale samples, characterized by X-ray diffraction (XRD) and scanning electron microscopy with energy-dispersive X-ray spectroscopy (SEM-EDS), mainly comprise quartz, dolomite, wurtzite, and calcite, with a minor amount of pyrite. Samples were cut from the same rock block into pieces with a size of 3 mm × 3 mm and polished to a depth of 200 μm. In the PDMS flow cell, only one side (3 mm in length and 200 μm in depth) of the shale piece contacted the fluid. The fracture channel width b was approximately 200 μm. The detailed fabrication process of the microfluidic chip is shown in Supporting Information Figure S5. (b) Microfluidic-visualization system. The dissolution and precipitation processes were recorded using a fluorescence microscope. Postexperiment, shale samples were removed from the flow cells and examined in detail with scanning electron microscopy−backscattered electron detection (SEM-BSE).

Under flowing conditions, the formation of precipitation zones becomes more complicated. The flow rate modifies the relative effect of advection to diffusion as the dominant transport mechanism, which eventually controls the location and amount of mineral precipitation.[30] Under extremely slow flow rates, barite precipitation primarily concentrates at the inlet of the flow channel at low pH, while a thin layer of precipitation covers the entire flow channel at high pH.[27] Xiong et al. reported that at high flow rates, a large volume of barite precipitates in larger fractures, with less precipitation in narrower sections.[31] Khormali and Ahmadi investigated the effect of flow rate on the amount of mineral precipitation, showing that lower flow rates result in more precipitation due to the longer residence time of the reactive fluid exposed to the solid surface.[32] These studies indicate diversity in preferential precipitation zones influenced by the pH and flow rate. However, most attention has been focused on the effect of either the pH or the flow rate in isolation. A more systematic investigation of the coupled effects of pH and flow rate is still lacking.

Microfluidics is a valuable experimental tool that, when combined with high-speed cameras or optical microscopes, allow real-time monitoring of fluid flow and reaction transport in two-dimensional geometries with spatial and temporal resolutions that outperform most core-flooding systems.[33−36] Poonoosamy et al. investigated the relationship between barite precipitation and pore size under various degrees of supersaturation using PDMS-fabricated pore networks with well-defined sizes and shapes.[37] However, traditional microfluidics fail to capture the complex fluid−rock interactions expected in natural rock systems.[38] To address this gap, Song et al. developed a micromodel method using real rock samples.[39] This approach integrates microfluidic channels etched into natural rock to visualize how acidic fluids interact with carbonate rocks, revealing dissolution rates and patterns. Similarly, Zhou et al. embedded actual rock samples in microfluidic models to simulate more realistic dissolution processes of soluble minerals.[40] Nooraiepour et al.'s microfluidic study on $CaCO_3$ synthesis in heterogeneous geological substrates demonstrating the random behavior of calcite crystal positioning and precipitation patterns influenced by nucleation probability.[42] Despite these advances, microfluidic studies of Barite precipitation have rarely incorporated real shale systems. Integrating actual shale samples into microfluidic models can provide deeper insights into the pore-scale mechanisms of barite precipitation under conditions that better reflect real-world scenarios.

Here, we aim to study barite precipitation under flowing conditions with various pH levels in fractured porous media. By combining microfluidic experiments and characterization tools such as XRD and SEM-EDS, we quantified the evolution of the alteration zones and observed two precipitation regimes. Furthermore, we defined a dimensionless number Da to elucidate the underlying mechanism for the two observed precipitation regimes. Our findings provide a basis for

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developing strategies to control precipitation formation and minimize pore blockage, which is critical for accurately modeling reactive transport in fractured porous media across diverse applications.

MATERIALS AND METHODS

We conducted flow-through precipitation experiments using a real-rock microfluidic visualization system (Figure 1 and Supporting Information Figure S1). The microfluidics were fabricated from shale samples obtained from a depth of 2400−2700 m in the Fengping oil field, Qinghai, China. X-ray diffraction (XRD) analysis revealed that the shale sample consists of 24.0% quartz, 47.1% dolomite, 6.1% wurtzite 4H, and 22.8% calcite with trace amounts of pyrite (Supporting Information Figure S2). To observe a clear alteration zone structure in a short time frame so that we can examine multiple conditions, we specifically selected a shale with a relatively high carbonate content. High-carbonate-content shale is widely used in various studies, including research by Li et al.[18,43] and Deng et al.[44,45] Energy-dispersive X-ray spectroscopy (EDS) images indicated a uniform elemental distribution (Supporting Information Figure S3). The porosity of the sampled core is 3.8467%, and the permeability k is 4.74×10^{-12} m^2; the pore size distribution of the shale is reported in Supporting Information Figure S4. The shale samples were cut into 3 mm × 3 mm pieces and polished to a thickness of 200 μm and then inserted into polydimethylsiloxane (PDMS) flow cells molded from silicon wafers.[40] Since the size of the shale chip was prepared to match precisely the fabricated PDMS cell, eliminating the need for glue. To further secure the setup, we reinforced the PDMS microfluidic chip with an acrylic frame. In these flow cells, only one side of the sample was exposed to the fluid, with the fracture channel width being approximately 200 μm (Supporting Information Figure S5).

We first injected HCl solution (0.01 mol/L, pH 2.0, Codow Standard titration solution) into the flow cell at a rate of 12 μL/min for 8.33 h. The HCl solution reacts with calcite in the rock, creating secondary porosity in the matrix bordering the channel, which will be termed alteration zones. Subsequently, we injected 0.06 mmol/L Na_2SO_4 and 2 mmol/L $BaCl_2$ solutions through two separate channels. The concentration of $BaCl_2$ is determined based on the Ba content in flowback water,[19] while the Na_2SO_4 concentration is such that the solution is mildly saturated with respect to barite (Table 1 and Supporting Information Table S1).[29] The total volume of solution injected at each stage is 6 mL. The pore volume V/V_0 will be used in the discussion instead of the injected solution volume (V) where V_0 is the fracture volume. The entire process was recorded using fluorescent microscopy visualization techniques (ZEISS Axiovert 5) coupled with a high-speed CMOS camera (Axiocam 208 color) (Figure 1b).[41,46] To distinguish between the liquid and solid phases, we introduced a fluorescent dye (0.1 mg/L, FLUORESCENCE SuperFluor488) into the solution. Supporting Information Figure S6 shows the solution preparation process. After the flowthrough experiments, the samples were dried and characterized using SEM-BSE and SEM-EDS (MIRA3 TESCAN), with an electron acceleration voltage of 20 kV; heavier minerals such as barite and pyrite are brighter in SEM-BSE images, while pore spaces are darker compared to silicates.[43] The experimental process and drying and cleaning procedures can be seen in Supporting Information Figures S7 and S8. We also analyzed some samples after removing the top 30 μm to verify the precipitation distribution (Supporting Information Figure S9); the details of the removing procedures are provided in Supporting Information Figure S10. Through image postprocessing (Supporting Information Figure S11), we calculated the average thickness of the alteration zone $d(t)$ and the distribution and amount of barite precipitation. Note that SEM images are 2D; while it is likely that the precipitation is not uniform in the vertical dimension, the area of the 2D precipitation distribution provides a proxy for the precipitate amount for semiquantitative analyses, as used in previous microfluidic precipitation studies.[36,47]

Experiments were conducted at four different flow rates (Q = 1.2, 4, 12, and 24 μL/min) and three pH levels (pH = 2, 8, and 11), with each condition repeated two times, totaling 24 experiments. All 24 samples were sourced from the same core and prepared following the same procedures outlined in Supporting Information Figure S5. They exhibit similar mineral compositions, with comparable proportions of mineral constituents. However, there is some mineralogical heterogeneity, leading to slight variations in the mineral content (Supporting Information Figures S2 and S3). The wide pH range ensured the observation of distinct nucleation behaviors. We quantified the interplay between advection and diffusion using the classic Péclet number ($Pe = vb/D$), where v is the average velocity in the fracture channel and $D = 9.37 \times 10^{-9}$ m^2/s is the molecular diffusion coefficient (Supporting Information Table S2).

Table 1. Reactions for the Minerals Present in the System[a]

mineral	reaction stoichiometry
quartz	quartz ⇌ SiO_2(aq)
calcite	calcite +2H^+ ⇌ Ca^{2+} + CO_2(aq) + H_2O
dolomite	dolomite +4H^+ → Ca^{2+} + Mg^{2+} + 2CO_2(aq) + 2H_2O
wurtzite	wurtzite+2H^+→H_2S(aq)+Zn^{2+}
pyrite	pyrite + H_2O + 3.5O_2(aq) ⇌ 2H^+ + 2SO_4^{2-} + Fe^{2+}
barite	Ba^{2+} + SO_4^{2-} ⇌ $BaSO_4$(s)

[a]However, given the timescale of the experiments, our discussion will focus on calcite and dolomite dissolution in the acid-flushing stage and barite precipitation in the second flooding stage.

RESULTS AND DISCUSSION

Evolution of the Alteration Zone. Figure 2 illustrates the evolution of the alteration zone thickness $d(t)$ under various experimental conditions. During the HCl flooding stage, at a constant flow rate (12 μL/min) and pH (pH 2) across all experiments, carbonate minerals such as calcite ($CaCO_3$) and dolomite ($CaMg(CO_3)_2$) dissolve, while the insoluble portions remain as a supporting frame. This heterogeneity in dissolution rates induces an alteration zone characterized by increased porosity and permeability. This zone can be clearly identified in the SEM images shown in Figure 2b and Supporting Information Figure S12. In the images recorded by the fluorescence microscope, the alteration zone appears as a shaded region (indicated by vertical arrows), and the dissolution front advances in a piston-like pattern. The thickness of the alteration zone $d(t)$ increases as a function of V/V_0, reaching approximately 120 μm after injecting 6 mL of solution; the evolution of the alteration zone with HCl flooding is shown in Supporting Information Movie S1.

In the subsequent $BaCl_2$ and Na_2SO_4 flooding stage, for cases with pH 2, the acid in the solution further reacts with carbonate minerals in the shale matrix. The alteration zone $d(t)$ is shown to increase continuously, and secondary alteration zones, denoted by d_{second}, are observed. From Figure 2a, it is evident that the end-of-run depths of the alteration zone decrease with the flow rate. At the end of the experiments, the thickness of the alteration zone is observed to be 480 μm at the smallest flow rate (Pe = 107), whereas this value remains about 160 μm at Pe = 2134. This negative correlation between the dissolution depth (alteration zone thickness) and flow rate suggests that the reactive flow process in our experiments is transport-limited. That is, the residence time of reactive fluid within the flow channel is the dominant transport mechanism. For the same value of V/V_0, curves for the slower flow rates correspond to a longer reaction time, thereby promoting deeper dissolution into the shale matrix. As the altered zone thickens, although the diffusion time of the solution from the fracture into the matrix increases,[44] the porosity in the alteration zone also increases,

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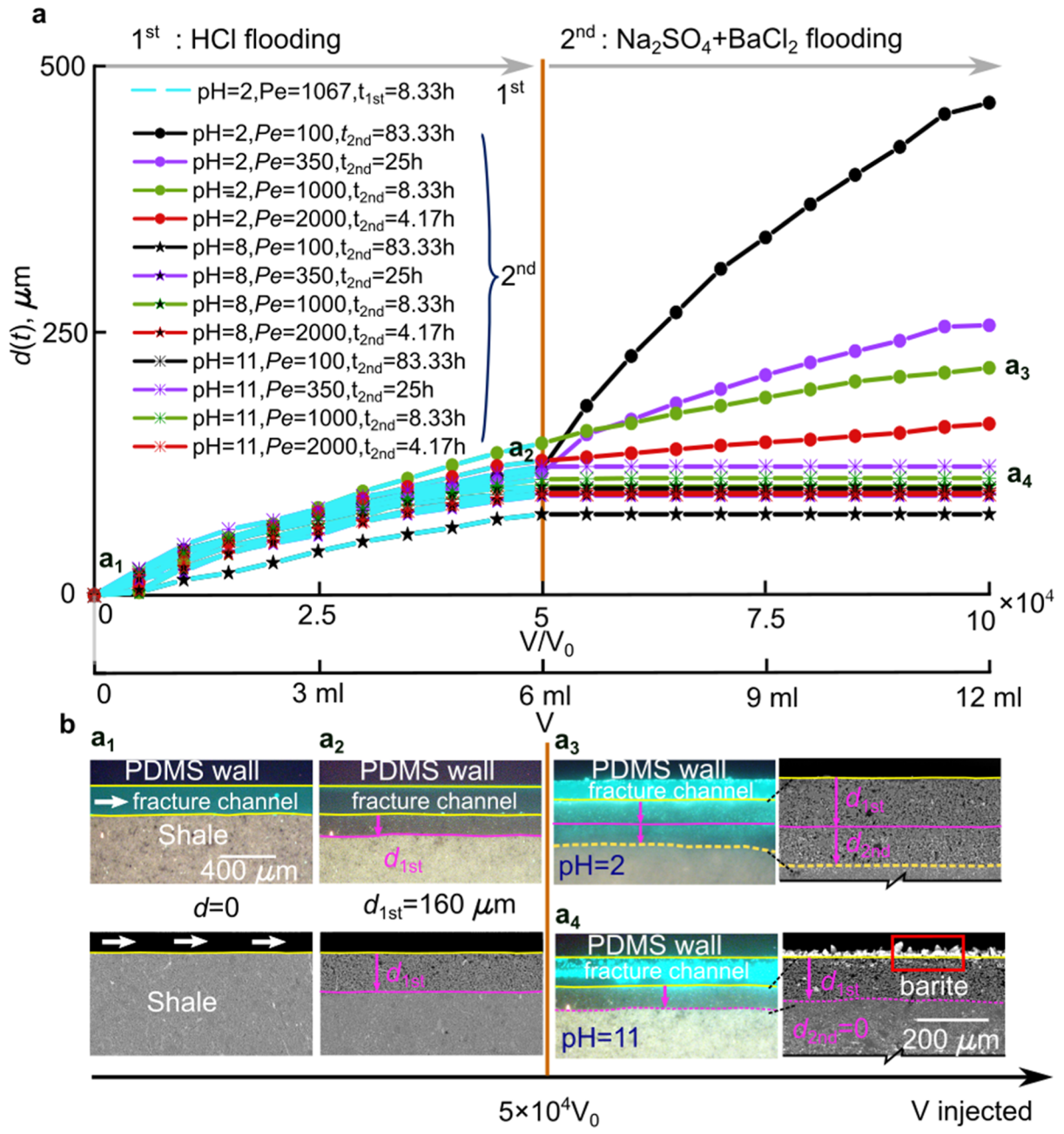


Figure 2. Observation and quantification of the alteration zone and barite precipitation. (a) During the HCl flooding at the same flow rate (12 μL/min) and the same pH (pH = 2) for all experiments, the alteration zone forms and its thickness $d(t)$ increases with flooding time, reaching approximately 120 μm at 5×10^4 V_0. In the subsequent stage with Na_2SO_4 + $BaCl_2$ flooding, $d(t)$ continues to increase for pH = 2 cases. The end-of-run depths of the alteration zone decreases with increasing flow rate (Pe), resulting in a larger d_{second} for lower Pe at 1×10^5 V_0. Lower flow rates allow longer residence time of HCl on the shale, promoting deeper dissolution into the shale matrix. (b) Imaging of the alteration zone and barite precipitation. (a_1–a_2). Fluorescence microscopy images show the evolution of the alteration zone during the first flooding stage. The gray-brown region represents the shale matrix, the alteration zone appears as a shaded region (indicated by vertical arrows), and the dissolution front advances in a piston-like pattern. This zone can also be clearly identified in the SEM images. (a_3). At the end of the experiment with pH = 2, pore-scale images illustrate the secondary alteration zone (d_{second}) compared to the initial flooding stage (d_{first}). SEM imaging reveals barite precipitation within this secondary alteration zone (a_4). For pH = 11, no dissolution occurs during the second flooding, resulting in d_{second} = 0. In this scenario, barite precipitation is observed on the shale surface. The two precipitation regimes with Na_2SO_4 and $BaCl_2$ flooding are also shown in Supporting Information Movies S2 and S3.

leading to an increased reaction surface area and transport of the reactive solution in this region. Therefore, the dissolution rate is not significantly affected. In contrast, at pH 8 or 11, no dissolution occurs during the second flooding stage, and the thicknesses of the alteration zones remain unchanged (d_{second} = 0). Additionally, we also observed a few white barite precipitations within the secondary alteration zone at pH = 2, while a considerable amount of precipitation scale formed on the shale surface at pH = 11. The two precipitation regimes will be discussed in Sections 3.2 and 3.3.

Two Precipitation Regimes. From the BSE and EDS images in Figure 3, two distinct precipitation regimes can be identified. In regime I (Figure 3a), substantial barite precipitation is observed on the fracture surfaces, as indicated

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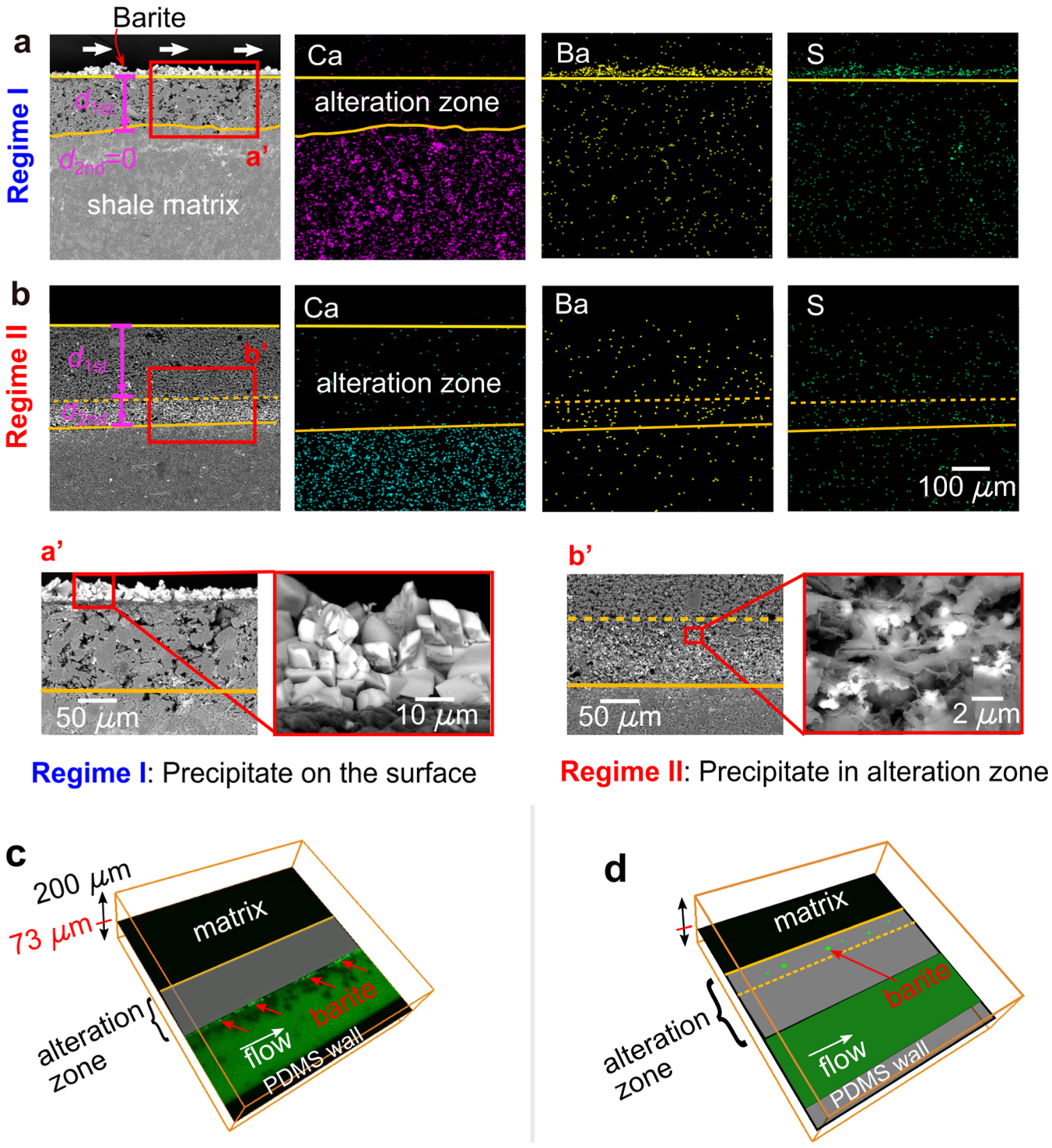


Figure 3. Observation of two precipitation regimes. (a) In regime I, a large amount of barite precipitation is observed with *Pe* = 107 and pH = 11 on shale surfaces by SEM-BSE images. These features of regime I is supported by the spatial distribution of elements (Ba, S) and the high-magnification SEM images (a′). According to the maps of various elements characterized by EDS, we can observe an obvious Ba element and S element gathered on the shale surface. Meanwhile, the extremely low content of Ca element confirms the alteration zone area. (b) In regime II, few barite precipitates are observed with *Pe* = 1067 and pH = 2 in the alteration zone formed during the second flooding. Ba element and S element are gathered but with the small amount in this secondary alteration zone. (c) Confocal microscopy images show significant precipitation-induced fluorescence accumulation on the shale wall (marked by red arrows) in the middle of the thickness direction (73 μm). (d) Due to the opacity of the shale, the precipitation distribution of the precipitation regime II cannot be observed by confocal microscopy. A schematic diagram showing that the precipitation is mainly distributed in the secondary alteration zone. We verified the precipitation distribution of regime II by SEM-BSE characterization after grinding 30 μm surface of the shale piece down (Supporting Information Figure S9).

by the enriched Ba and S on the shale surface. Meanwhile, compared to the matrix, the alteration zone is depleted of Ca, indicating the dissolution of calcium minerals in this region. In regime II (Figure 3b), only a few barite precipitates are observed

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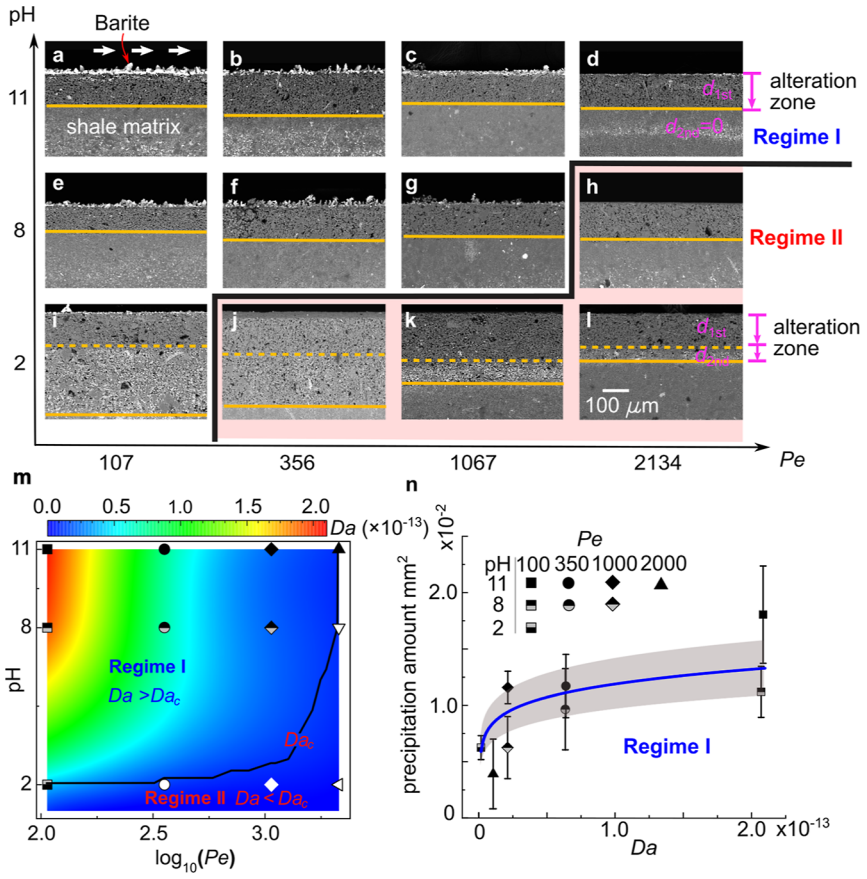


Figure 4. (a–l) Experimental phase diagram of precipitation regimes (I and II). The *Pe* number (or the flow rate *v*) increases from the left to right. The pH increases from the bottom to the top. As the flow rate increases and pH decreases, the precipitation regime shifts from I [(a–g,i) precipitation on the shale surface] to II [(h,j–l) precipitation in the alternation zone]. (m) The transition of precipitation regimes is controlled by the dimensionless number *Da*. A threshold of *Da* can be obtained as $Da_c \approx 1.05 \times 10^{-14}$. For $Da > Da_c$, barite precipitates on the shale surface (regime I), while barite precipitation is observed in the alternation zone in the case of $Da < Da_c$. (n) In regime I, the amount of barite precipitation can be well corrected with *Da* using a power-law function $a_{pre} = Da^{0.148\pm0.003}$, indicating that increasing *Da* can promote the precipitation on the surface. The shadow denotes the 95% confidence interval.

in the alteration zone formed during the second flooding. Due to the small amount of precipitation, the Ba and S signals in the secondary alteration zone are not very strong. Similar precipitation regimes were observed in benchtop experiments by Li et al.[18] under nonflowing conditions, where the precipitation regime depended on the carbonate content within the shale. However, our experiments show that the precipitation regimes are influenced by both the flow rate and pH.

To understand the morphology of precipitated barite in the two regimes, we used SEM to capture high-resolution images of the precipitation zones (Figure 3a,b). In regime I (*Pe* = 107 and pH = 11), precipitation particles on the shale surface are large and have well-defined shapes (Figure 3a′), whereas in regime II (*Pe* = 1067 and pH = 2), particles in the alteration zone are smaller and poorly defined in shape (Figure 3b′). This appears to be different from that in previous studies. For example, Ruiz-Agudo et al.[28] reported that barite particles tend to become more equidimensional and smaller as pH increases. The difference may be attributed to constraints imposed by pore size as when barite precipitates within the secondary alteration zone, confined spaces and limited solution volumes naturally restrict the particle size.[25,48,49] While their experiments were conducted in a fluid cell, equal volumes (1 mL) of $BaCl_2$ and Na_2SO_4 solutions with varying pH values were mixed.

Given that SEM images only capture 2D features, we repeated a subset of the experiments using confocal microscopy to further understand the distribution of precipitation in 3D. For precipitation regime I, images show significant precipitation-induced fluorescence accumulation on the shale surface at a depth of 73 μm (Figure 3c), and the dynamic growth can be found in Supporting Information Movie S4. For regime II, due to the opacity of the shale, the precipitation distribution could not be observed by confocal microscopy directly. We verified the precipitation distribution of regime II by SEM-BSE characterization after removing the top 30 μm of the shale sample (Supporting Information Figure S9). We found that the precipitate band on the matrix surface disappeared after polishing, although a small amount of barite precipitate remained in the alteration zone. The amount of precipitate was much less than that on the unpolished shale surface, as shown in Supporting Information Figure S9e–h. We show that the precipitate is mainly distributed in the secondary alteration zone (Figure 3d).

Precipitation Regimes Controlled by the Nucleation Rate and Advection Rate. The experimental phase diagram of precipitation regimes in Figure 4 summarizes the effects of flow rate and pH, highlighting regime I (Figure 4a–g,i) where barite precipitation is observed on the fracture surfaces and regime II (Figure 4h,j–l) where the precipitation is mainly distributed in

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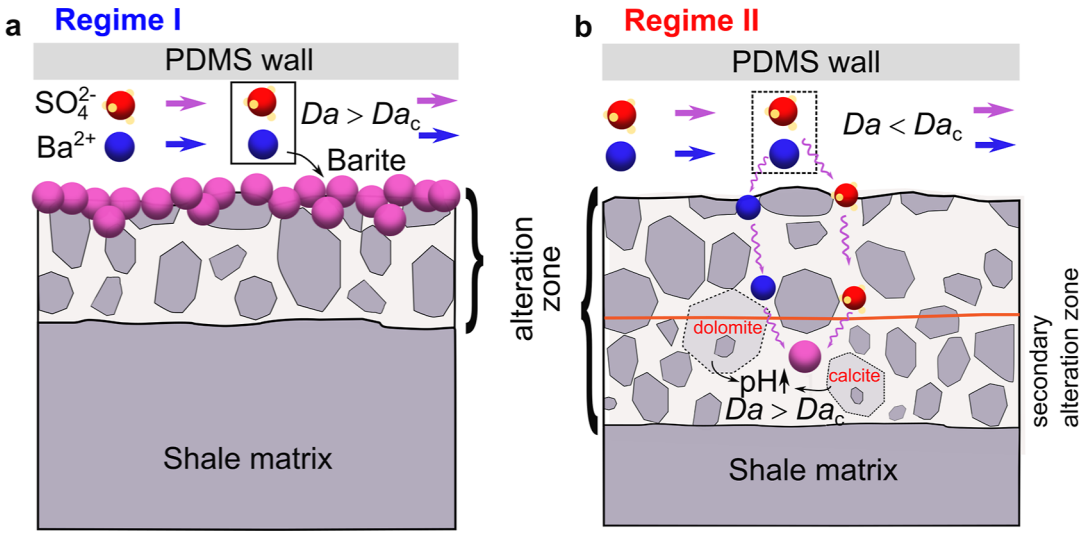


Figure 5. Mechanism of two precipitation regimes. (a) In regime I with a large *Da* number, which means faster precipitation rate, slower flow rate, and longer experimental time. Barite forms on the surface because it precipitates rapidly before diffusing inward, especially under conditions of high pH (where precipitation is much faster). In the experiment with pH = 2 and *Pe* = 107, while the precipitation rate is relatively slower, barite precipitation on the fracture surface is observed because the long reaction time allowed a small amount of precipitation to accumulate on the surface. (b) In regime II with a low *Da* number, the precipitation rate is slower (at low pH) and the flow rate is faster and barite forms within the alteration zone. When the reactive fluid with Ba^{2+} and SO_4^{2-} at pH 2 diffuses to the calcite front, it reacts with carbonate minerals in the shale, increasing local pH and promoting barite precipitation in the alteration zone.

the secondary alteration zone. As observed from Figure 4a–l, as the flow rate increases and pH decreases, the mass of precipitated barite on the shale surface appears to decrease, and precipitation shifts from regime I to regime II. Classical nucleation theory (CNT) is usually used to describe the barite precipitation process. According to CNT, changing the solution pH alters supersaturation (Ω) in our study, leading to different nucleation rates. To elucidate the underlying mechanism of the observed precipitation regimes and as inspired by Deng et al.,[50] we introduce a modified Damkohler number *Da* to compare the precipitation rates with advection rates; the nucleation rates are used as an approximation of the precipitation rates.

$$Da = \frac{J'}{v} \tag{1}$$

In eq 1, J' is the nucleation rate in unit of m/s, calculated using the eqs S1–S5 in the Supporting Information Text S1. v is the advection rate [m/s]. Since $Pe = vb/D$, advection rate v can also be expressed as $v = PeD/b$. Substituting $v = PeD/b$ into eq 1, the *Da* can be written as

$$Da = \frac{J'b}{PeD} \tag{2}$$

The dimensionless number *Da* allows us to correlate the flow rate and pH with precipitation regimes effectively. As shown in Figure 4m, *Da* decreases with higher flow rates or a lower pH, leading to a transition in precipitation regimes. We identify a critical Da_c in our experiments, above which precipitation occurs on the fracture surface (regime I), and below which precipitation occurs within the matrix (regime II). The upper left corner on the phase diagram in Figure 4 corresponds to the large *Da*, which means faster precipitation rate, slower flow rate, and longer experimental time. Barite forms on the surface because precipitation occurs rapidly before diffusion into the matrix, especially under high pH conditions (where precipitation is much faster), as shown in Figure 5a. In the experiment with pH 2 and *Pe* = 107, while the precipitation rate is relatively slower, barite precipitation on the fracture surface is observed because the long reaction time allowed a small amount of precipitation to accumulate on the surface. At a low *Da* number (the down right corner on the phase diagram in Figure 4), the precipitation rate is slower (at low pH)[51] and the flow rate is faster and barite forms within the alteration zone. When the reactive fluid with Ba^{2+} and SO_4^{2-} at pH 2 diffuses to the calcite front, it reacts with carbonate minerals in the shale, increasing local pH and promoting barite precipitation in the alteration zone (Figure 5b). However, due to the relatively small volume of the reactive fluid, only a few precipitates form before the dissolution front recedes. This results in localized precipitates rather than continuous deposition. As shown in Figure 4i–l, as the altered zone thickens, the increased porosity allows more solution to enter the matrix, meaning more reaction fluid can interact with the minerals in the matrix, providing more reactive surface area and expanding the reaction zone.

To improve quantitative understanding of how the relationship between precipitation and advection controls barite precipitation on the rock surface, we performed image processing (Supporting Information Figure S11) to extract and calculate the amount of barite precipitation (a_{pre}) for regime I. Here, a_{pre} was calculated as the area of precipitated barite normalized by A_s. As shown in Figure 4n, a_{pre} correlates well with *Da*, following a power-law function $a_{pre} = Da^{\lambda}$ with $\lambda = 0.148 \pm 0.003$. Increasing the *Da* promotes the formation of a barite layer on the shale surface. Taken into account the probabilistic nature of nucleation,[42] the locations of precipitation accumulation are not fixed when comparing Figures 4 and S13 under the same experimental conditions. In some places, the amount of precipitation is particularly high. Because once a layer of barite forms, the probability of nucleation on the preformed barite will be greatly increased given that the effective surface tension between barite layers is 22.8 mJ/m^2,[28] significantly lower than that between the barite and the rock surface. This lower energy barrier on barite surfaces results in a higher precipitation rate[52]

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(Supporting Information eqs S1−S4). In other words, once the first layer of barite forms on the rock surface, this positive feedback promotes overall barite nucleation and precipitation for a larger *Da*.

CONCLUSIONS

This study systematically investigates the coupled effects of the pH and flow rate on the dynamics of barite precipitation in fractured porous media. Through real-rock microfluidic experiments, we observed two distinct precipitation regimes: precipitation on the fracture surface (regime I) and within the alteration zone (regime II). The study introduced the dimensionless numbers *Da* as a critical parameter for predicting these regimes, revealing that higher flow rates and lower pH favor regime II, while lower flow rates and higher pH favor regime I. These findings enhance our understanding of the pore-scale mechanisms that control mineral precipitation in subsurface environments. If the precipitating mineral is changed, the nucleation and precipitation dynamics will differ due to variations in properties like solubility, crystal structure, and interaction with the shale matrix. However, nucleation behavior is a common feature across mineral systems and is influenced by factors such as the pH and flow rate. While the precipitation behavior may vary between minerals, our study provides a useful reference for understanding how nucleation rate and advection rate influence precipitation regimes. This insight is broadly applicable to other mineral precipitation systems, even though the extent of the effect may differ. We acknowledge that although dimensionless numbers can provide useful insights when certain assumptions are satisfied, they may not fully capture the complexity of time-dependent factors in precipitation dynamics.

By elucidating the conditions that control the occurrences of the two precipitation regimes and the amount of precipitation, our findings provide actionable knowledge for mitigating environmental impacts and optimizing resource management practices involving mineral precipitation. For contaminant remediation, by controlling precipitation regimes to maximize precipitation on the surface of the contaminant medium, mineral precipitation forms an in situ sealing layer that minimize the release of the pollutants from the matrix.[53,54] Further replacing the cap with adsorptive materials can achieve the immobilization of pollutants.[54] Coprecipitation is also an important mechanism for contaminant remediation.[55,56] By controlling the relative relationship between the nucleation rate and advection rate, we can increase the precipitation amount and thereby enhance contaminant remediation efficiency. Moreover, the location of the precipitation will affect the accessibility for subsequent reaction and future redissolution and release; as Deng et al. found that with the increase of alteration zone thickness, mineral dissolution becomes more and more limited,[44] the precipitation in the alteration zone is less likely to be redissolved and released. In shale gas production, based on the occurrence of the two precipitation regimes, it is possible to minimize pore blockage by controlling the chemical composition and flow rates of injected fluids and thus to manage permeability and mass transfer and to reduce the frequency of fracturing events for conserving water and other resources.[12] In addition, it should be noted that when regulating the pH of the injected solution to control its precipitation regime, the influence of pH on the properties of scale inhibitors should be considered to achieve the best scale inhibition effect.[57,58] More importantly, the introduction of the dimensionless number Da as a parameter for predicting precipitation regimes at the microscale may help enhance the accuracy of macroscale reactive transport models. These models are essential for anticipating the long-term behavior of subsurface formations under various operational conditions. This can guide future research and help develop sustainable extraction strategies and the design of in situ pollutant remediation schemes that minimize environmental impacts.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.langmuir.4c03532.

Nucleation rate calculation procedure; macroscopic petrophysical properties of the sampled core calculation procedure; details of the experimental system; mineral composition characterized by XRD and EDS; pore size distribution of the shale; fabrication of real-rock microfluidics; solution preparation steps; flowchart of the experimental protocol; drying and cleaning procedures after the experiments ; BSE images comparing whether the surface is polished to verify if the precipitation is only distributed on the surface of the shale; procedure for removing the top 30 μm of shale samples; image postprocessing; comparison of SE and BSE images: pore structure in the matrix and alteration zone; phase diagram of repeated experiments; images of precipitation distribution in the fracture channel; BSE and SE images of the alteration zone area; equilibrium reactions during the experiment; summary of precipitation experiments; and parameter values for barite nucleation using classical nucleation theory (PDF)

Dissolution with the HCl flooding (*Pe* = 1067, pH = 2) (AVI)

Precipitation with Na_2SO_4 and $BaCl_2$ flooding in regime I (*Pe* = 1067, pH = 11) (AVI)

Precipitation with Na_2SO_4 and $BaCl_2$ flooding in regime II (*Pe* = 1067, pH = 2) (AVI)

Precipitation growth process (regime I) observed by confocal microscopy (*Pe* = 1067, pH = 11) (AVI)

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Notes

The authors declare no competing financial interest.

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